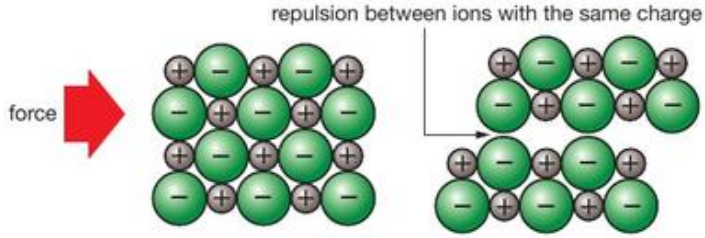
Ionic crystals tend to be brittle. This is because any small distortion of a crystal will bring ions with the same charge alongside each other. Like charges repel and so the crystal splits itself apart.



▲ Figure 7.20 Ionic crystals tend to be brittle

KEY POINT

Organic solvents include ethanol (alcohol) and hydrocarbons, such as hexane and those found in petrol. If you are interested in these, you could explore the organic chemistry section of this book (Unit 4).

Ionic substances tend to be soluble in water.

Ionic compounds tend to be insoluble in organic solvents.

EXTENSION WORK

The reasons that ionic compounds tend to be soluble in water are quite complicated. Water is a covalent molecule (Chapter 8) but the electrons in the bonds are more attracted towards the oxygen end of the bond. This makes the oxygen slightly negative and the hydrogen slightly positive – the molecule is called **polar**. This means that reasonably strong forces can be formed between water molecules and ions, which provide the energy to break the lattice apart.

Not all ionic substances are soluble in water: magnesium oxide isn't soluble in water because the attractions between the water molecules and the ions aren't strong enough to overcome the very strong electrostatic forces of attraction between magnesium and oxide ions.

Hexane is non-polar and does not form strong enough attractions to the ions to break apart the ionic lattice.

THE ELECTRICAL CONDUCTIVITY OF IONIC SUBSTANCES

HINT

Molten just means that the salt has been melted – it is a liquid.

Ionic compounds don't conduct electricity when they are solid because the *ions* are fixed in position and are not free to move around. They do, however, conduct electricity when they are **molten** (have melted) or if they are dissolved in water (in aqueous solution). This happens because the *ions* then become free to move around. It is really important that you use the correct words when explaining this. Do not use the word 'electrons'. You must talk about the **ions** being free to move.

CHAPTER QUESTIONS

SKILLS INTERPRETATION 7

SKILLS CRITICAL THINKING 8

1 a Explain what is meant by i an ion and ii ionic bonding.

b In each of the following cases, write down the electronic configurations of the original atoms and then explain (in words or diagrams) what happens when:

i sodium bonds with chlorine to make sodium chloride

ii lithium bonds with oxygen to make lithium oxide

iii magnesium bonds with fluorine to make magnesium fluoride.

SKILLS INTERPRETATION

2 Draw dot-and-cross diagrams to show the ions formed (outer electrons only) when:

a potassium combines with fluorine

b calcium combines with bromine

c magnesium combines with iodine.